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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Cloud Storage Technologies

Object, Block, and File Storage Compared

SIMATS ENGINEERING · CSE DEPARTMENT

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What Is Cloud Storage?

Definition

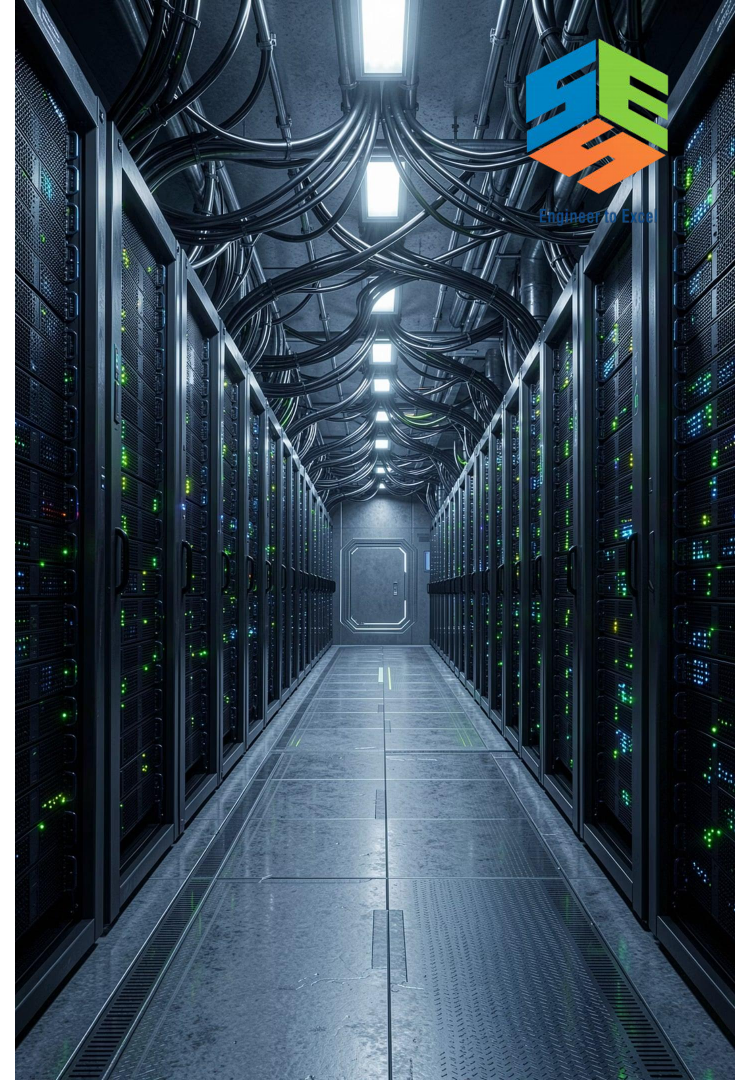
Remote data storage accessed over the internet, hosted on distributed server infrastructure.

Why It Matters

Eliminates local hardware limits.
Enables scalability, redundancy, and global access.

Three Core Types

Object Storage · Block Storage · File Storage — each optimized for different workloads.



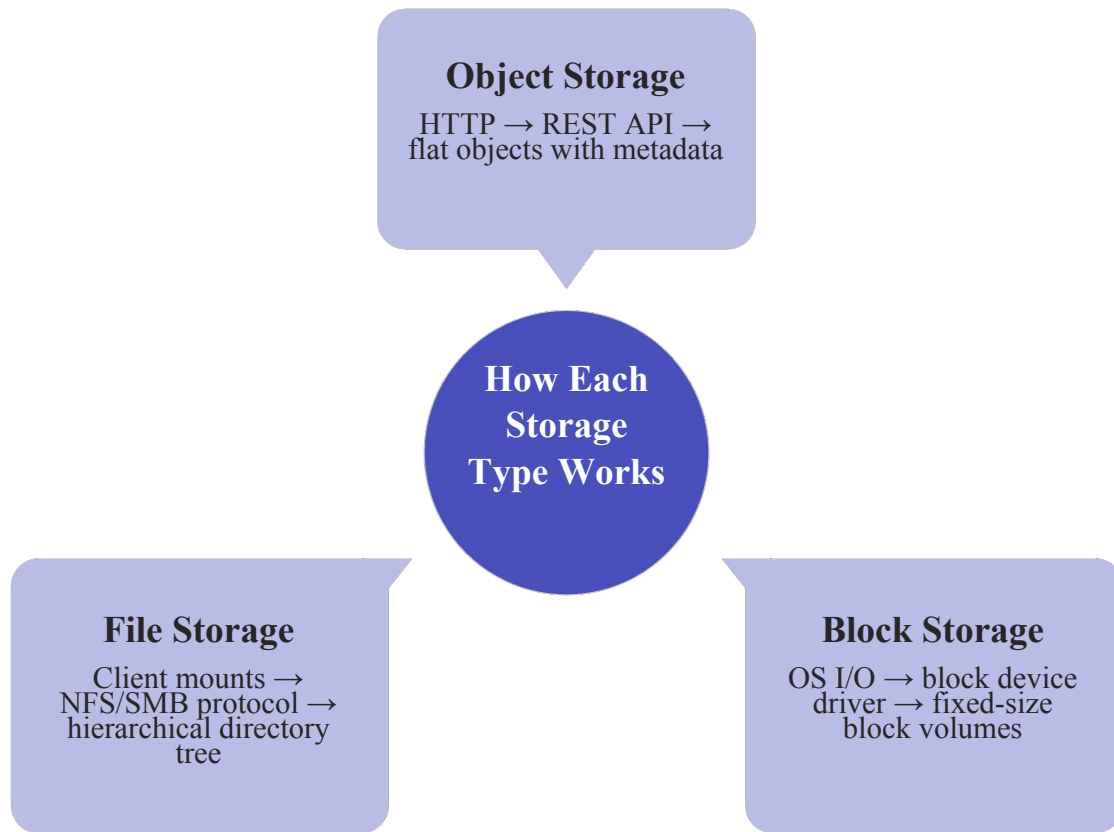


Definitions & Comparison

Attribute	Object Storage	Block Storage	File Storage
Structure	Flat namespace, metadata-rich objects	Fixed-size raw data blocks	Hierarchical folders & files
Access	HTTP / REST API	iSCSI / Fibre Channel	NFS / SMB protocols
Best For	Unstructured data, media, backups	Databases, OS volumes	Shared team file systems
Scalability	Virtually unlimited	Limited, must pre-provision	Moderate



How Each Storage Type Works

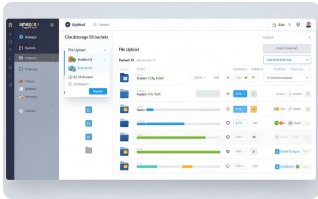


Each storage type exposes a different interface to the operating system or application — from raw block-level I/O to high-level file paths to API-driven object retrieval.



Engineer to Excel

Industry Relevance & Practical Examples



Amazon S3 — Object

Stores billions of images, videos, and backups for Netflix, Airbnb, and NASA. Pay-per-use, 99.999999999% durability.



AWS EBS — Block

Provides persistent SSD volumes for EC2 instances running Oracle and MySQL databases. Low-latency, high IOPS workloads.



Google Drive / NAS — File

Shared file systems for university labs, enterprise teams, and media production houses. Familiar folder-based access.

Key Takeaways

Match Storage to Workload

No single type wins — choose based on access pattern, scalability need, and latency requirement.

Object = Scale, Block = Speed, File = Share

This mental model covers 90% of real-world cloud storage decisions.

Industry Moves to Hybrid

Modern architectures often combine all three types within a single application stack.



For exams: Remember S3 = Object · EBS = Block · NFS/Google Drive = File



THANKYOU